

ROOP SAI CHARAN KOTHAMASU

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EDUCATION

University of Florida

Master of Science in Computer Science

Gainesville, FL

Aug 2025 – Aug 2027

SRM University

Bachelor of Technology in Computer Science

Andhra Pradesh, India

Mar 2021 – Mar 2025

TECHNICAL SKILLS

Languages: JavaScript, Python, Java, C, C++, SQL, Solidity

Front-end Technologies: React.js, Next.js, HTML5, CSS3, Tailwind CSS, Bootstrap

Back-end Technologies: Node.js, Express.js, REST APIs, MongoDB, PostgreSQL, MySQL

Cloud and DevOps: AWS (EC2, CloudWatch, S3), CI/CD, Docker, Git, Linux/UNIX

Machine Learning: TensorFlow, Keras, PyTorch, Scikit-learn, CNN, RNN, LSTM, Transfer Learning

Data Science Tools: Pandas, NumPy, Matplotlib, OpenCV, Jupyter Notebook

Development Tools: GitHub, VS Code, Google Colab, Postman, Agile/Scrum

EXPERIENCE

SRM University AP

Special Project Intern – Blockchain Research

Andhra Pradesh, India

Dec 2024 – Mar 2025

- Designed and deployed Solidity smart contracts on Ethereum for tamper-proof patient data storage, integrating Node.js REST APIs and MongoDB for off-chain data management across 500+ simulated users.
- Implemented JWT-based role access control for 3 distinct user roles, eliminating unauthorized access entirely and ensuring 99.9% data integrity throughout the system.
- Collaborated directly with supervising professor to iterate on system architecture, rapidly adapting to blockchain concepts and delivering a fully functional prototype within a 3-month timeline, contributing to an ongoing publication and patent filing.

APSSDC

AWS Cloud Computing Intern

Remote

Jun 2023 – Aug 2023

- Launched scalable web applications on AWS EC2 with automated monitoring, improving system uptime by 55% and reducing downtime incidents by 40%.
- Engineered CI/CD pipelines using AWS CodePipeline and GitHub Actions, reducing manual deployment time by 70% and accelerating release cycles from 5 days to 1.5 days.
- Optimized resource allocation across 12+ EC2 instances using CloudWatch analytics, decreasing infrastructure costs by 25% while maintaining 99.9% availability.

SRM University AP

Machine Learning Research Intern

Andhra Pradesh, India

Jan 2022 – Mar 2022

- Developed end-to-end ML pipelines processing 15,000+ records with 95% accuracy using Python, Pandas, and Scikit-learn.
- Implemented ensemble learning techniques (Random Forest, XGBoost) improving prediction stability by 85% and F1-score from 0.72 to 0.89.
- Conducted model evaluation using cross-validation, achieving ROC-AUC of 0.92+ across 6 different ML algorithms.

PROJECTS

MeetMint: AI-Powered Productivity Dashboard | Go, React, Gemini 1.5 Flash, RAG, SQLite

March 2026 – May 2026

- Architected a full-stack AI dashboard using Go and React to transform meeting audio into structured project metadata, leveraging Gemini 1.5 Flash for low-latency transcription analysis and task extraction.
- Engineered a local Retrieval-Augmented Generation (RAG) pipeline using SQLite and embeddings, enabling context-aware Q&A over meeting history with support for manual transcript refinement.
- Developed a premium Glassmorphism UI with Framer Motion and secure multimedia BLOB storage, maintaining a robust test suite of 35+ backend and integration tests.

Decentralized Patient Record Management System | Node.js, MongoDB, Solidity, Ethereum

Dec 2024 – Mar 2025

- Architected blockchain-based healthcare platform using Solidity smart contracts and Node.js REST APIs, enabling secure management for 500+ simulated users with 99.9% data integrity.
- Enforced granular permission management across doctor, patient, and admin roles using JWT authentication, achieving full compliance with healthcare data privacy standards.
- Released Ethereum smart contracts with audit logging, decreasing manual verification by 40% through immutable transaction records.

Music Streaming Platform | Next.js, React, Tailwind CSS, Stripe, Supabase

Jul 2023 – Sep 2023

- Built full-stack subscription platform using Next.js and React, serving 200+ active users with Stripe payment integration.
- Improved performance through SSR and caching, reducing page load time by 25% (3.2s to 2.4s) and improving Core Web Vitals.
- Designed PostgreSQL database in Supabase managing 1,500+ songs with 95% query efficiency and responsive UI achieving 100% mobile compatibility.

PUBLICATIONS

R.S. Charan et al. (2024). *Analyzing Lossless Image Compression Techniques for IoT Devices*, ICEEICT 2024.

R.S. Charan et al. (2024). *An Explainable AI Driven Hybrid Model for Enhanced Intrusion Detection in Network Security*.

R.S. Charan et al. (2024). *Decentralized Patient Record Management System* — Publication and patent under process.